Nonlinear Relationships

Beyond the Straight Line

ECON3500: Econometrics and Applications

Spring 2026

Roadmap

[TOC]

What You'll Learn Today

Central Question

What if the relationship between X and Y is **not** a straight line?

By the end of this chapter, you will:

Model **polynomial relationships** (quadratic, cubic)

Compute and interpret **marginal effects**

Estimate **interaction effects** between variables

Use **logarithmic transformations** effectively

Test functional form with Ramsey RESET

Apply these tools in Stata

Why Nonlinear?

The Linear Assumption

So far, we've assumed relationships are linear:

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

Interpretation: A one-unit increase in X changes Y by β_1 units, **regardless of X's current value**.

But is this realistic?

Does the first year of education have the same effect as the 20th?

Does doubling advertising from $1K to $2K have the same effect as doubling from $100K to $200K?

Does the effect of training depend on whether the worker is already skilled?

Real-World Examples of Nonlinearity

Diminishing returns:

Fertilizer on crop yield (eventually toxic!)

Study hours on test scores (exhaustion sets in)

Experience on wages (plateaus after 30+ years)

Increasing returns:

Network effects (social media value)

Learning curves (skills compound)

Interactions:

Effect of education may depend on ability

Effect of experience may differ by gender

Three Tools for Nonlinearity

Polynomial regression — Add X^2, X^3, etc. to model curvature

Interaction terms — Include $X_1 \cdot X_2$ to allow effects to vary

Logarithmic transformations — Use $\log(Y)$ or $\log(X)$ for percentage changes

> **Key Insight**
>
> All three remain **linear in parameters** — we can still use OLS!

Polynomial Regression

The Quadratic Model

Quadratic Regression Model

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + u_i$$

Why quadratic?

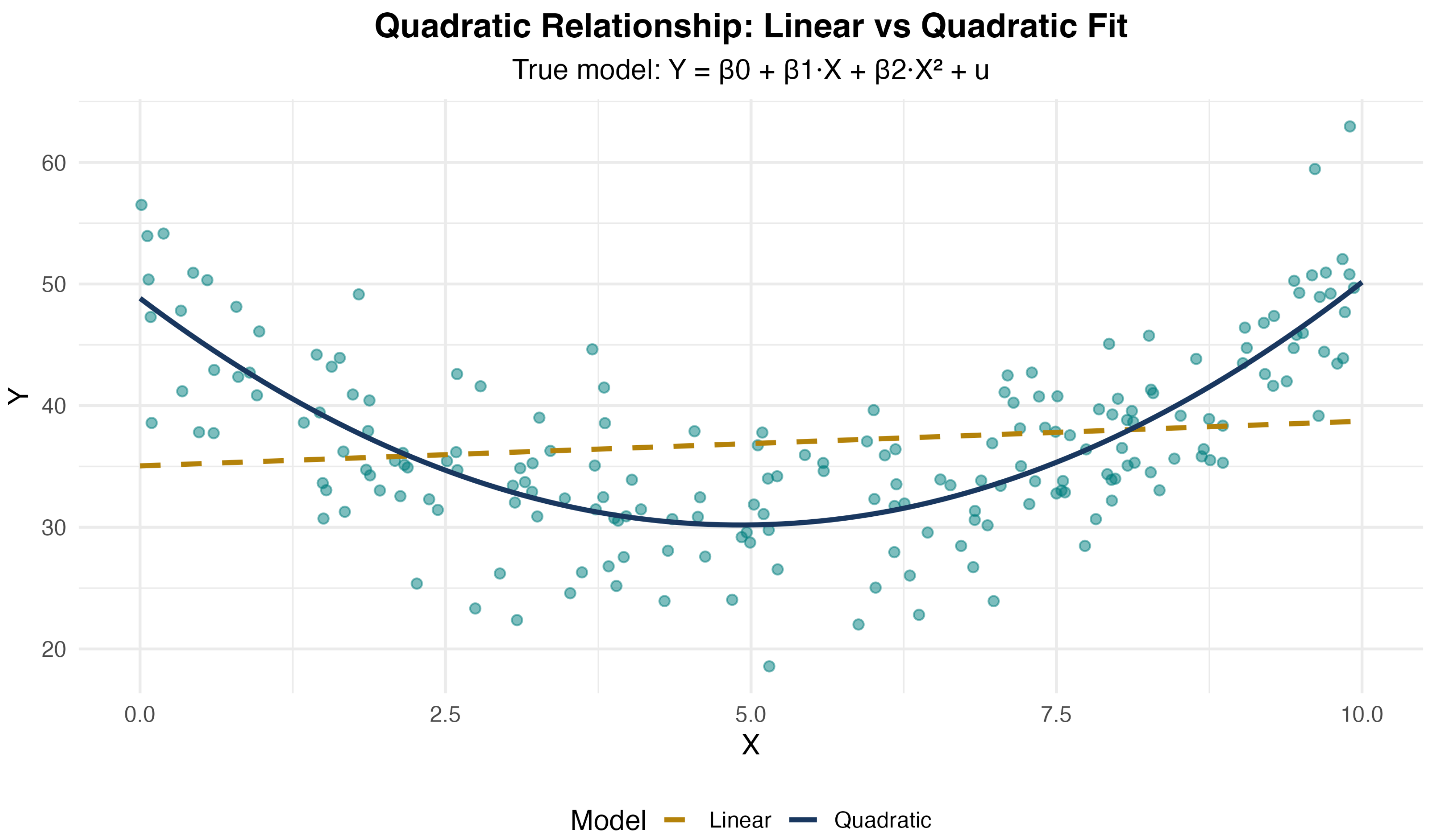
Allows for **curvature** in the relationship

Can capture diminishing or increasing returns

Flexible, but not too flexible (avoids overfitting)

Still linear in parameters: We estimate $\beta_0, \beta_1, \beta_2$ via OLS!

Visualizing Quadratic Relationships



Interpreting Quadratic Coefficients

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + u_i$$

Key point: The effect of X on Y is **no longer constant**!

Marginal Effect in Quadratic Model

$$\frac{\partial Y}{\partial X} = \beta_1 + 2\beta_2 X$$

The marginal effect **depends on the value of X**!

Interpretation:

At $X = 0$: marginal effect is β_1

At $X = 10$: marginal effect is $\beta_1 + 20\beta_2$

The marginal effect changes by $2\beta_2$ for each unit increase in X

Example: Earnings and Experience

Model:

$$\log(wage) = \beta_0 + \beta_1 \cdot experience + \beta_2 \cdot experience^2 + u$$

Estimated coefficients:

```
log(wage) = 1.5 + 0.08*experience − 0.0012*experience^2
```

Interpretation:

$\hat{\beta_1} = 0.08 > 0$: wages initially increase with experience

$\hat{\beta_2} = -0.0012 < 0$: inverted-U shape (diminishing returns)

Marginal effect = $0.08 - 2(0.0012) \times experience$

Calculating Marginal Effects

Estimated model:

$$\log(wage) = 1.5 + 0.08 \cdot exper - 0.0012 \cdot exper^2$$

At $exper = 10$ years:

$$\frac{\partial \log(wage)}{\partial exper} = 0.08 - 2(0.0012)(10) = 0.08 - 0.024 = 0.056$$

At $exper = 30$ years:

$$\frac{\partial \log(wage)}{\partial exper} = 0.08 - 2(0.0012)(30) = 0.08 - 0.072 = 0.008$$

Interpretation: The wage boost from an additional year of experience falls from 5.6% (at 10 years) to 0.8% (at 30 years).

Finding the Turning Point

For an inverted-U relationship, where does the maximum occur?

Set the marginal effect to zero:

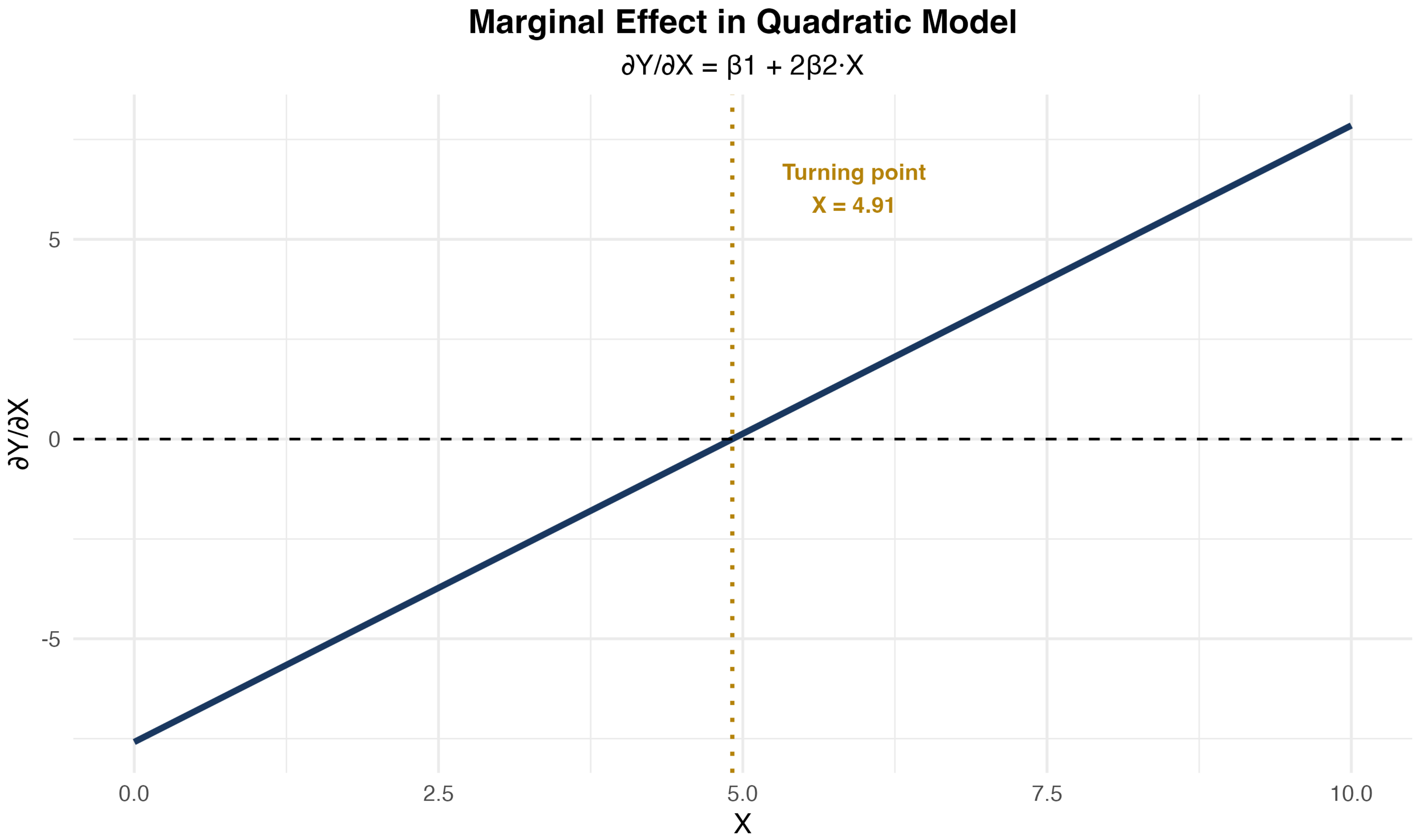
$$\beta_1 + 2\beta_2 X^* = 0 \quad \Rightarrow \quad X^* = -\frac{\beta_1}{2\beta_2}$$

Example: $\log(wage) = 1.5 + 0.08 \cdot exper - 0.0012 \cdot exper^2$

$$exper^* = -\frac{0.08}{2(-0.0012)} = \frac{0.08}{0.0024} \approx 33.3 \text{ years}$$

Interpretation: Wages peak at about 33 years of experience, then decline.

Visualizing Marginal Effects



Knowledge Check: Quadratic Model

Question

You estimate: $sales = 100 + 15 \cdot price - 0.5 \cdot price^2$

1. What is the marginal effect of price on sales at $price = 10$?
2. At what price are sales maximized?

Answer

Part 1: Marginal effect $= 15 - 2(0.5)(10) = 15 - 10 = 5$. At \$10, raising price by \$1 increases sales by 5 units.

Part 2: Turning point: $price^* = -\frac{15}{2(-0.5)} = 15$. Sales are maximized at $price = 15$.

Cubic and Higher-Order Polynomials

Cubic model:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \beta_3 X_i^3 + u_i$$

Marginal effect:

$$\frac{\partial Y}{\partial X} = \beta_1 + 2\beta_2 X + 3\beta_3 X^2$$

When to use: Multiple turning points, S-shaped patterns.

Warning: Higher-order polynomials can **overfit** and behave erratically at extremes. Use sparingly!

Stata Implementation: Polynomial Regression

Create polynomial terms:

```
generate experience2 = experience^2
generate experience3 = experience^3
```

Estimate model:

```
regress log_wage education experience experience2, robust
```

Compute marginal effect at mean experience:

```
summarize experience
scalar exper_mean = r(mean)
display "Marginal effect = " _b[experience] + 2*_b[experience2]*exper_mean
```

Shortcut using margins:

```
margins, dydx(experience) at(experience=10)
```

Interaction Effects

What Are Interactions?

Interaction Effect

The effect of X_1 on Y **depends on** the value of X_2.

Examples:

Effect of education on wages may differ by gender

Effect of advertising may depend on product quality

Effect of fertilizer may depend on rainfall

Mathematical representation:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 (X_{1i} \times X_{2i}) + u_i$$

The **interaction term** is $X_1 \times X_2$.

Continuous × Continuous Interaction

Model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 (X_{1i} \cdot X_{2i}) + u_i$$

Marginal effect of X_1:

$$\frac{\partial Y}{\partial X_1} = \beta_1 + \beta_3 X_2$$

Interpretation:

When $X_2 = 0$: effect of X_1 is β_1

When $X_2 = 10$: effect of X_1 is $\beta_1 + 10\beta_3$

The effect of X_1 changes by β_3 for each unit increase in X_2

Example: Education and Experience Interaction

Question: Does the return to education depend on experience?

Model:

$$wage = \beta_0 + \beta_1 \cdot education + \beta_2 \cdot experience + \beta_3(education \times experience) + u$$

Estimated:

```
wage = 2.0 + 1.5*education + 0.3*experience + 0.05*(education*experience)
```

Interpretation:

For someone with 0 years experience: return to education = $1.50/year

For someone with 10 years experience: return = $1.50 + 0.05(10) = $2.00/year

Education becomes more valuable as you gain experience!

Visualizing Continuous × Continuous Interactions

Binary × Continuous Interaction

Model:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 D_i + \beta_3 (X_i \cdot D_i) + u_i$$

where D_i is a dummy variable (0 or 1).

Interpretation:

When $D = 0$: $Y = \beta_0 + \beta_1 X$ (baseline group)

When $D = 1$: $Y = (\beta_0 + \beta_2) + (\beta_1 + \beta_3)X$ (treatment group)

β_2: intercept shift | β_3: slope shift

Example: Gender Wage Gap

Model:

$$wage = \beta_0 + \beta_1 \cdot education + \beta_2 \cdot female + \beta_3(education \times female) + u$$

Estimated:

```
wage = 3.0 + 2.0*education - 1.5*female - 0.3*(education*female)
```

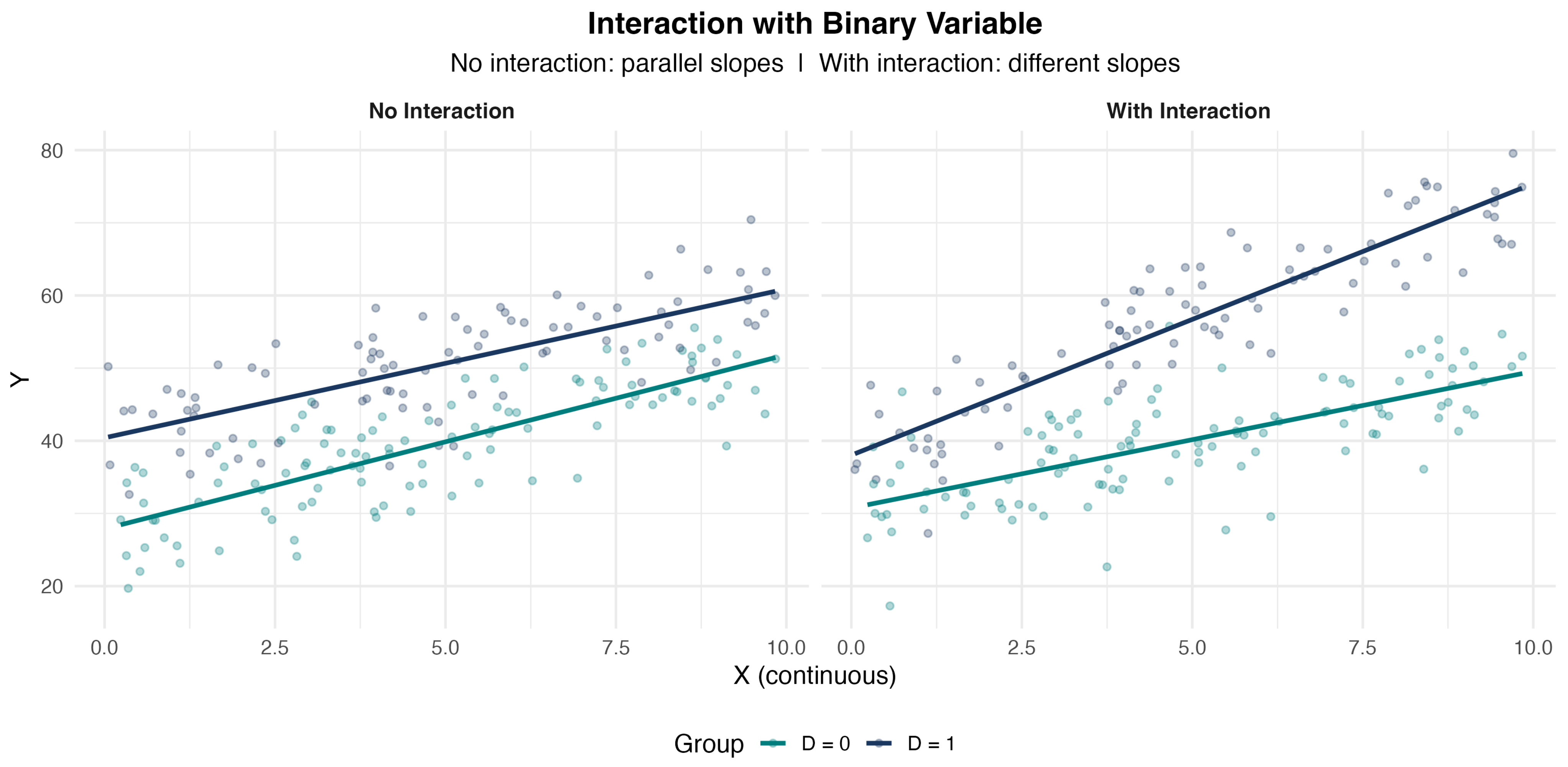
Interpretation:

Men ($female = 0$): $wage = 3.0 + 2.0 \cdot education$

Women ($female = 1$): $wage = 1.5 + 1.7 \cdot education$

Women earn $1.50 less at 0 education; $0.30 less per year of education. Gap widens with education!

Visualizing Binary × Continuous Interactions



Parallel lines ⇒ no interaction. Different slopes ⇒ interaction present!

Knowledge Check: Interaction Effects

Question

You estimate: $price = 50 + 10 \cdot quality + 5 \cdot advertising + 2 \cdot (quality \times advertising)$

1. What is the marginal effect of advertising when quality = 5?
2. What is the marginal effect of advertising when quality = 10?

Answer

Marginal effect of advertising = $5 + 2 \times quality$

Part 1: At quality = 5: effect = $5 + 2(5) = 15$

Part 2: At quality = 10: effect = $5 + 2(10) = 25$

Advertising is more effective for higher-quality products!

Stata Implementation: Interactions

Method 1: Create interaction manually

```
generate educ_exper = education * experience
regress wage education experience educ_exper, robust
```

Method 2: Factor variable notation

```
regress wage c.education##c.experience, robust
```

For dummy × continuous:

```
regress wage c.education##i.female, robust
```

Compute marginal effects at different values:

```
margins, dydx(education) at(experience=(0 10 20 30))
```

Common Mistakes with Interactions

Including interaction but not main effects — Always include X_1 and X_2 when you include $X_1 \times X_2$!

Interpreting β_1 as "the" effect of X_1 — With interactions, the effect depends on other variables; β_1 is only the effect when $X_2 = 0$.

Forgetting to center variables — Centering makes β_1 interpretable as effect at average X_2.

Testing wrong hypothesis — Test $H_0 : \beta_3 = 0$ to check if interaction is significant.

Logarithmic Specifications

Why Logarithms?

Three key advantages:

Percentage interpretation — Coefficients represent **percentage changes**, not unit changes

Diminishing returns automatically — $\log(X)$ is concave (built-in curvature!)

Elasticities — Direct estimates of elasticity in log-log models

Important: Can only take log of **positive** variables!

Four Logarithmic Models

Model	Equation	Interpretation
Level-level	$Y = \beta_0 + \beta_1 X$	$\Delta Y = \beta_1 \Delta X$
Log-level	$\log Y = \beta_0 + \beta_1 X$	$\% \Delta Y = 100 \beta_1 \Delta X$
Level-log	$Y = \beta_0 + \beta_1 \log X$	$\Delta Y = (\beta_1 / 100) \% \Delta X$
Log-log	$\log Y = \beta_0 + \beta_1 \log X$	$\% \Delta Y = \beta_1 \% \Delta X$

Key insight: Each log transforms a unit change into a percentage change!

Level-Level (Baseline)

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

Interpretation: A one-unit increase in X is associated with a β_1-unit change in Y.

Example: $wage = 5 + 2.5 \cdot education$ — Each additional year of education increases wages by $2.50/hour.

Log-Level

$$\log(Y_i) = \beta_0 + \beta_1 X_i + u_i$$

Log-Level Interpretation

A one-unit increase in X is associated with a $100\beta_1$ % change in Y.

Example: $\log(wage) = 1.5 + 0.08 \cdot education$ — Each additional year of education increases wages by 8%.

Level-Log

$$Y_i = \beta_0 + \beta_1 \log(X_i) + u_i$$

Level-Log Interpretation

A 1% increase in X is associated with a $\beta_1/100$ unit change in Y.

Example: $wage = 2 + 5\log(experience)$ — A 1% increase in experience increases wages by \$0.05/hour. Doubling experience (100% increase) raises wages by \$5/hour.

Log-Log (Elasticity)

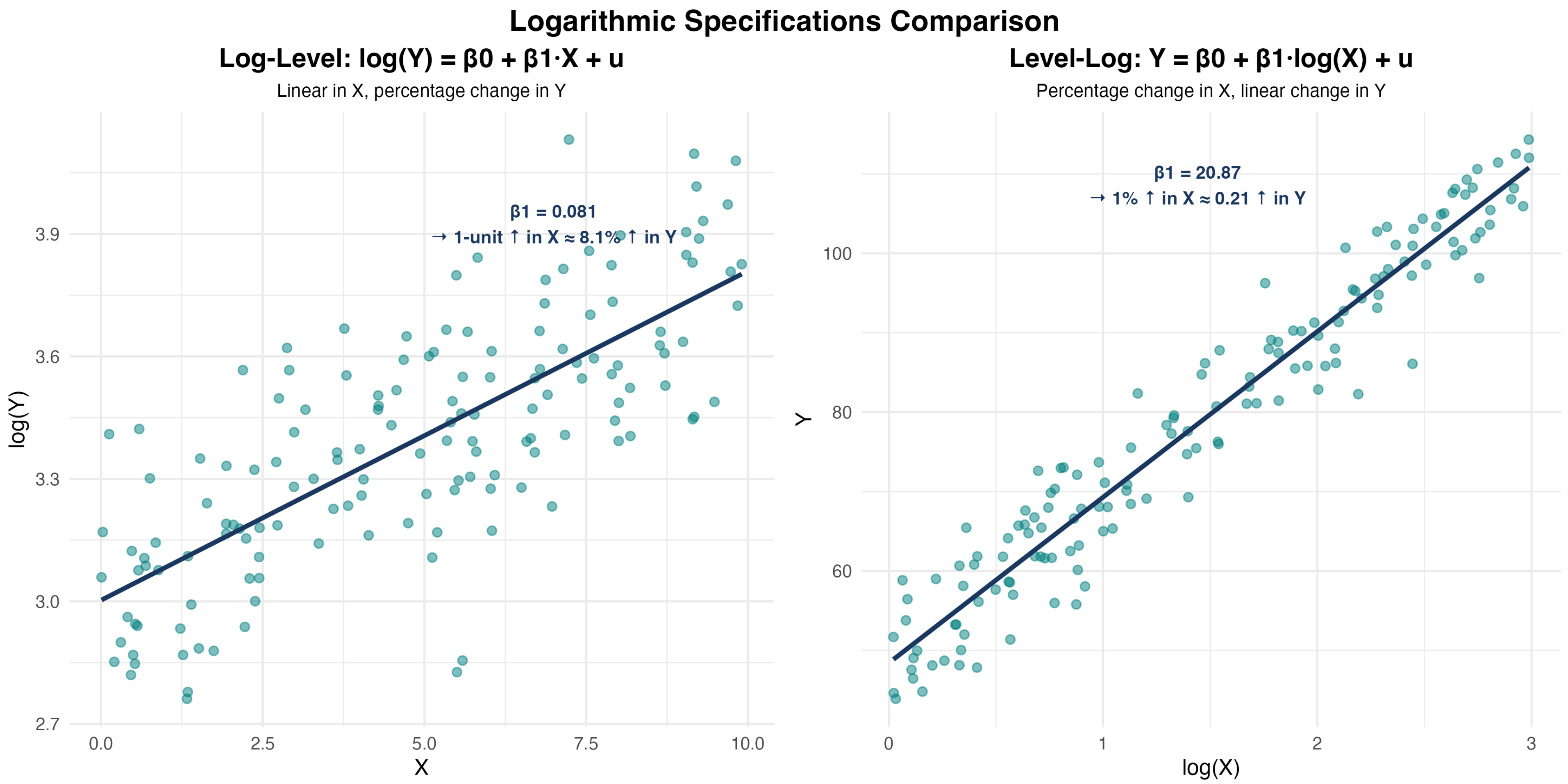
$$\log(Y_i) = \beta_0 + \beta_1 \log(X_i) + u_i$$

Log-Log Interpretation

β_1 is the **elasticity** of Y with respect to X. A 1% increase in X leads to a β_1% change in Y.

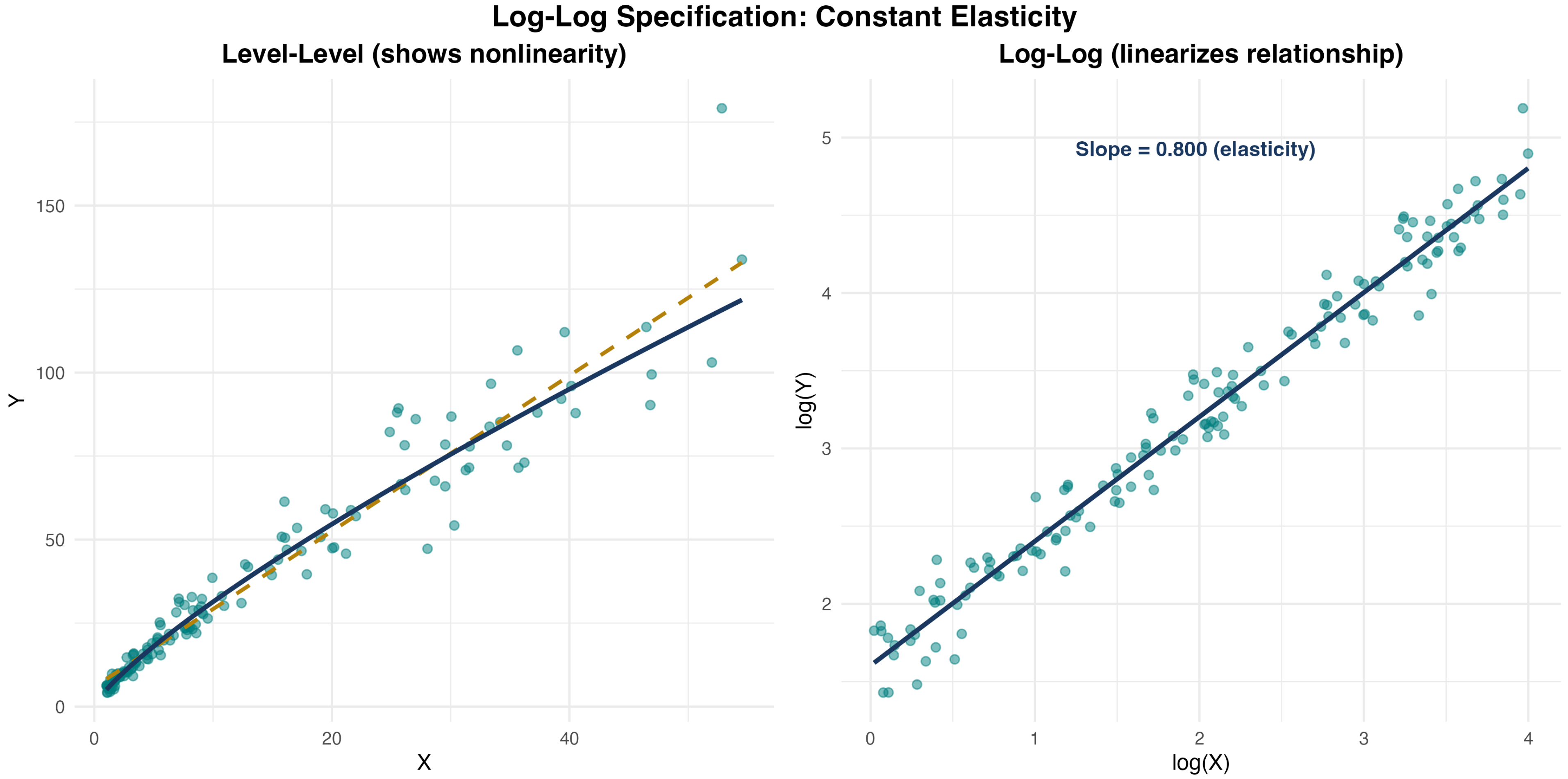
Example: $\log(quantity) = 2.5 - 1.2 \log(price)$ — A 1% increase in price leads to a 1.2% decrease in quantity (price elasticity of demand).

Visualizing Log Specifications



$\log$ transformations create natural curvature

Log-Log Specification in Detail



On log-log scale, relationship is linear with slope = elasticity

Knowledge Check: Log Specifications

Question

Match each estimated model to its interpretation:

a. $wage = 5 + 0.3 \cdot experience$

b. $\log(wage) = 1.2 + 0.04 \cdot experience$

c. $wage = 8 + 2.5\log(experience)$

d. $\log(wage) = 1.5 + 0.15\log(experience)$

e. Each year of experience raises wages by $0.30

ii. Each year of experience raises wages by 4%

iii. A 1% increase in experience raises wages by 0.15%

iv. Doubling experience raises wages by $2.50

Answer

a. → i) (level-level), b) → ii) (log-level), c) → iv) (level-log), d) → iii) (log-log)

Interpretation Guide Summary

Stata Implementation: Logs

Generate log variables:

```
generate log_wage = log(wage)
generate log_sales = log(sales)
generate log_price = log(price)
```

Estimate different specifications:

```
* Log-level
regress log_wage education experience, robust

* Level-log
regress wage log_experience log_sales, robust

* Log-log
regress log_quantity log_price log_income, robust
```

Important: Always use natural log (`log` in Stata), not log10 or log2!

Testing Functional Form

How Do We Know Which Model to Use?

We have many choices: Linear vs quadratic vs cubic? Logs or levels? Interactions or no interactions?

Three approaches:

Theory — What does economic theory suggest?

Visualization — Plot the data and residuals

Formal tests — Ramsey RESET test

The Ramsey RESET Test

Ramsey RESET (Regression Specification Error Test)

Tests whether nonlinear functions of fitted values help explain Y. Detects **omitted variables** or **incorrect functional form**.

Procedure:

Estimate original model, obtain fitted values $\hat{Y}$

Estimate augmented model: $Y = \beta_0 + \beta_1 X + \gamma_1 \hat{Y}^2 + \gamma_2 \hat{Y}^3 + u$

Test $H_0 : \gamma_1 = \gamma_2 = 0$ (F-test)

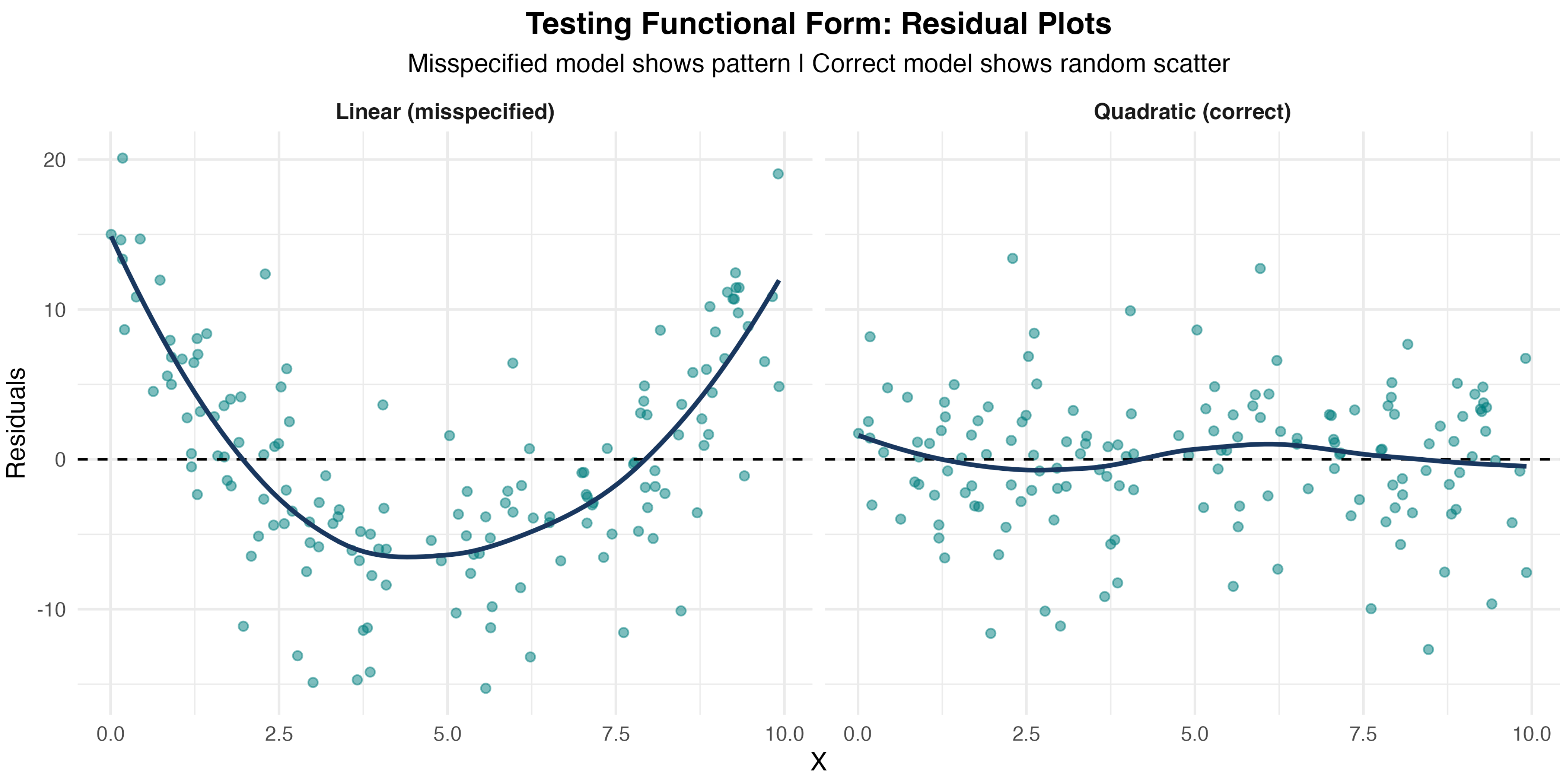
If reject H_0: functional form is misspecified

Interpreting RESET Results

If we reject H_0 (p-value < 0.05): Current model is misspecified; try adding polynomial terms, logs, or interactions; re-run RESET after adjusting.

If we fail to reject H_0: No strong evidence against current specification (but doesn't prove it's correct!).

Visualizing Functional Form Tests



Good fit: residuals randomly scattered. Bad fit: systematic pattern

Stata Implementation: RESET Test

```
* Estimate model
regress wage education experience, robust

* Run RESET test
estat ovtest
```

Example output: Ramsey RESET test using powers of fitted values; H_0: Model has no omitted variables; if Prob > F = 0.0000, we reject — try adding quadratic terms or logs!

Bringing It All Together

The Nonlinear Toolkit

Tool	Use When	Gives You
Polynomial	Curved relationships, turning points	Marginal effects that vary with X
Interactions	Effect depends on another variable	Different slopes for different groups
Logarithms	Wide ranges, percentage interpretation	Elasticities, diminishing returns

Key principle: All remain **linear in parameters** — we can use OLS and all our inference tools!

Key Formulas Summary

Model	Marginal Effect
Quadratic	$\frac{\partial Y}{\partial X} = \beta_1 + 2\beta_2 X$
Interaction	$\frac{\partial Y}{\partial X_1} = \beta_1 + \beta_3 X_2$
Log-level	$\frac{\%\Delta Y}{\Delta X} = 100\beta_1$
Level-log	$\frac{\Delta Y}{\%\Delta X} = \beta_1/100$
Log-log (elasticity)	$\frac{\%\Delta Y}{\%\Delta X} = \beta_1$

Turning point: $X^* = -\frac{\beta_1}{2\beta_2}$ (quadratic model)

Looking Ahead

What we've learned: How to model curved relationships (polynomials), varying effects (interactions), and percentage/elasticity interpretation (logs); how to test and choose functional forms.

Next chapter: Hypothesis Testing and Confidence Intervals — formal inference, t-tests and F-tests, confidence intervals, joint hypothesis tests.

The real world is nonlinear — now you can model it!